Care Bot

Elderly Medication Companion

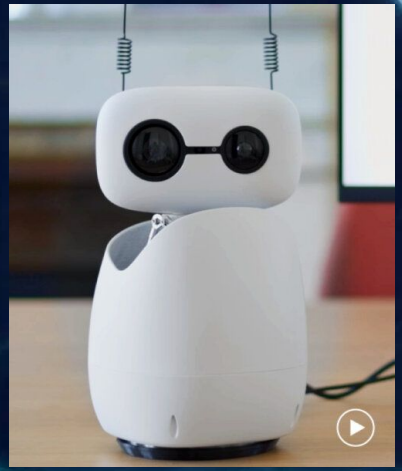
The Need for Automated Care

The Crisis: Why Medication Verification Matters

The Challenge

Medication **Non-Adherence** & **Complexity** Lead to Avoidable Harm.

Our Solution: CareBot



Autonomous AI agent visually confirms medication intake.

The Future of Automated Care: Benefits & Impact

Improved Patient Outcomes

Proactive monitoring and **personalized treatment plans** lead to better health results.

Scalable Healthcare Solutions

Efficient resource management and wider access to care through automation.

System Architecture & Data Flow

Jetson Orin 8Gb

- Reachy Daemon
- Core Reachy Agent Loop

Brev Cloud: NVIDIA L40S (48 GiB)

- NVIDIA-Nemotron-Nano-VL-12B-V2-FP8

Google Sheets

- Medication Times and Dosages

MinMax

- Text to Speech then Streaming back to the Agent Loop

User Persona Profile: The Care Circle

PRIMARY USER: Margaret, 74

Conditions: Diabetes, Hypertension

Meds: Multiple Daily

Meds

Lives Alone

Lives Alone

Low Tech Comfort

Pain Point: Confusion & Skipped Doses

Needs: Simple, Friendly Reminders

SECONDARY USER: Sarah, 46 (Daughter, Remote Caregiver)

Role: Remote Caregiver

Process: Daily Check-ins

Pain Point: Constant Anxiety

Needs: Reliable Remote Visibility

TERTIARY USER: Dr. Chen, 52 (Prescribing Physician)

Role: Primary Care Physician

Pain Point: Unreliable Self-Reports

Needs: Verifiable Adherence Data

Demo Session Flow

1. Alert & Verify

Announce & Verify Med

2. Confirm & Log

Confirm Intake & Wellness

3. Notify Caregiver

Update Dashboard

Differentiating Features

Intake Verification

Visual Intake Confirmation

Private Symptom Logging

Secure On-Device

Natural Language Config

Voice Configuration

Alerts & Tracking

Proactive Monitoring

Challenges

Jetson Model Size

Model was too large to run well on the Jetson.

Llama Version Issues

Ran into problems having the right version of Llama for Jetson.

Hardware Failures

Bad hardware meant debugging microphone failures on the robot.

Audio Architecture

Switched from Agora to local agent loop and Minmax for audio control.

Connectivity & Lag

Slow internet forced relocation to use ethernet and minimize delays.

Future Roadmap

Out of scope for the hackathon, but part of the natural product roadmap.

- Multi-patient support
- Vitals integration
- Pharmacy integration
- Multilingual support
- Drug interaction checking
- Clinical trial integration